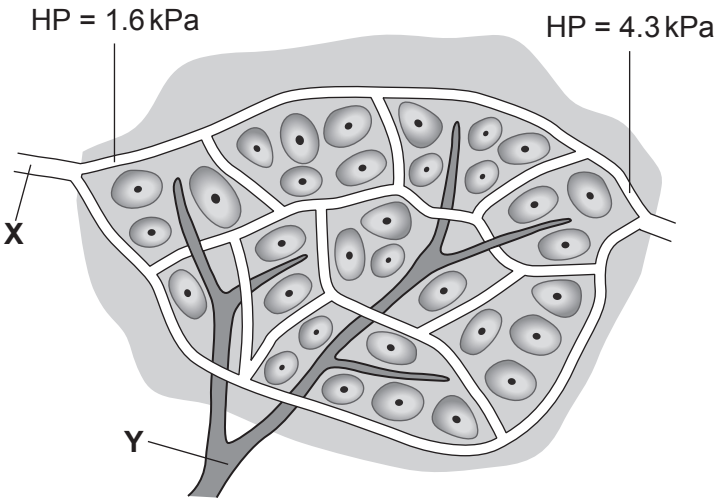


2. **Image 2.1** shows a diagram of the blood supply to the cells of a body tissue.

Image 2.1



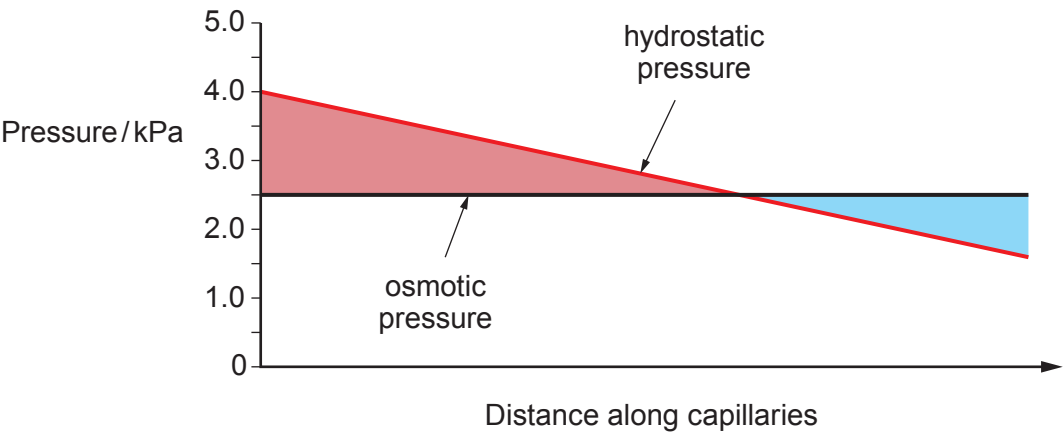
HP = hydrostatic pressure

- (a) (i) Identify the structures labelled **X** and **Y** in **Image 2.1**. [2]
- X**
- Y**
- (ii) **Draw an arrow** on **Image 2.1** to show the direction of blood flow through the capillary network. [1]



(b) **Graph 2.2** shows the changes in the hydrostatic pressure and osmotic pressure of the blood flowing through the capillaries.

Graph 2.2



(i) Explain why the hydrostatic pressure decreases as the blood travels along the capillaries. [2]

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(ii) Explain why the osmotic pressure remains constant as the blood travels along the capillaries. [2]

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(iii) Use the information provided in **Graph 2.2** to explain why structure **Y** in **Image 2.1** is necessary. [2]

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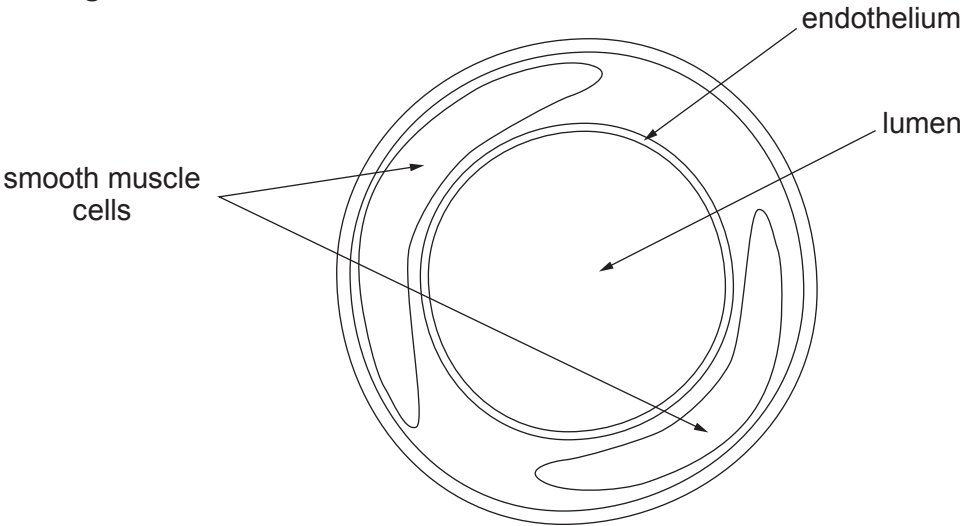
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- (c) Arterioles are responsible for the regulation of blood flow through a capillary network. **Image 2.3** shows a section through an arteriole.

Image 2.3



Explain how the structure of the wall of the arteriole, shown in **Image 2.3**, can reduce blood flow to a capillary network. [2]

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- (d) Arterioles are also responsible for the distribution of blood around the body. **Table 2.4** shows the blood flow in the human body at rest and during vigorous exercise.

Table 2.4

Body structure	Blood flow / cm ³ min ⁻¹	
	At rest	During vigorous exercise
heart	250	750
kidneys	1200	600
skeletal muscles	1000	125 000
skin	400	1900
stomach & intestines	1400	600
brain	750	750
other	600	400
total	5600	130 000

- (i) Calculate the percentage increase in blood flow to the skin during vigorous exercise.

[2]

Percentage =

- (ii) Suggest the significance of increased blood flow to the skin during exercise.

[1]

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